***Solution:***

Find the midpoint of the class intervals and tabulate it.

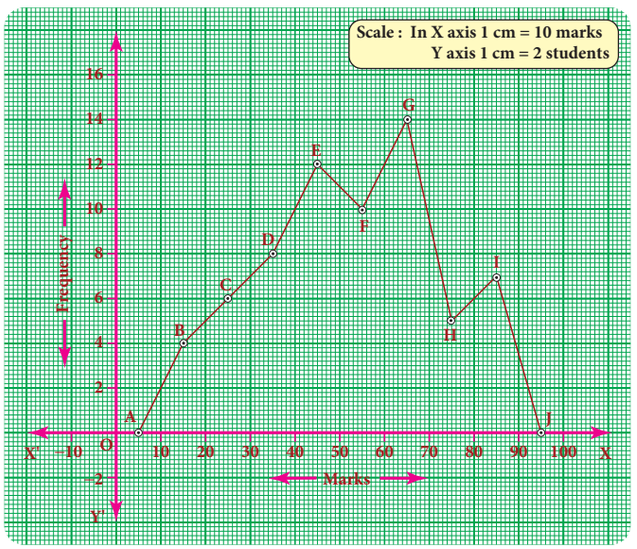
| Class interval ( C.I) | Mid point (x) | Frequency (f) |
|---|---|---|
| 10 – 20 | 15 | 4 |
| 20 – 30 | 25 | 6 |
| 30 – 40 | 35 | 8 |
| 40 – 50 | 45 | 12 |
| 50 – 60 | 55 | 10 |
| 60 – 70 | 65 | 14 |
| 70 – 80 | 75 | 5 |
| 80 – 90 | 85 | 7 |

The points are A(5,0) B(15,4) C(25,6) D(35,8) E(45,12) F(55,10) G(65,14) H(75,5) I(85,7) J(95,0).

In the graph sheet, mark the midpoints along the x- axis and the frequency along the y- axis.

We take the imagined class as 0 – 10 at the beginning and 90 – 100 at the end , each with frequency 'zero'.

From the table, plot the points. We draw the line segments AB, BC, CD, DE, EF, FG, GH, HI, IJ to obtain the required frequency polygon ABCDEFGHIJ.




## Exercise 6.2

1. Which of the following data can be represented in a histogram?
   (i) The number of mountain climbers in the age group 20 to 60 in TamilNadu.
   (ii) Production of cycles in different years.
   (iii) The number of students in each class of a school.
   (iv) The number votes polled from 7 am to 6 pm in an election.
   (v) The wickets fallen from 1 over to 50th over in a one day cricket match.

2. **Fill in the blanks:**
   (i) The total area of the histogram is _____ to the total frequency of the given data.
   (ii) A graph that displays data that changes continuously over the periods of time is _____.
   (iii) Histogram is a graphical representation of _____ data.



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